

Wuao Liu

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 [Google Scholar](#)
 [GitHub](#)

Education

- 08/2024 – Present **Ph.D. in Computer Science**, *University of Massachusetts Amherst*, Amherst, MA
- 08/2021 – 04/2023 **M.S. in Robotics**, *University of Michigan*, Ann Arbor, MI
- 09/2017 – 06/2021 **B.Eng. in Robotics**, *Zhejiang University*, Hangzhou, China

Publications

- [1] **Wuao Liu**, Mustafa Chasmai, Subhansu Maji, Grant Van Horn. *Masked Autoencoders with Limited Data: Does It Work? A Fine-Grained Bioacoustics Case Study*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop (FGVC13), 2026. **Oral Presentation** [\[paper\]](#)
- [2] Logan Lawrence, Oindrila Saha, Rangel Daroya, Mustafa Chasmai, **Wuao Liu**, Max Hamilton, Aaron Sun, Seoyun Jeong, Fabien Delattre, Subhansu Maji, Grant Van Horn. *RealBirdID: Benchmarking Bird Species Identification in the Era of MLLMs*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. [\[paper\]](#)
- [3] Mustafa Chasmai, **Wuao Liu**, Subhansu Maji, Grant Van Horn. *Bioacoustic Geolocation: Species Sounds as Geographic Signals*. International Conference on Machine Learning (ICML), 2026. [\[paper\]](#)
- [4] Jike Zhong, Yuxiang Lai, Ming Li, Yuheng Li, **Wuao Liu**, Behzad Dariush, Konstantinos Psounis, Shao-Yuan Lo. *From Shortcuts to Reasoning: Robust Post-Training of Theory of Mind with Reinforcement Learning*. International Conference on Machine Learning (ICML), 2026. [\[paper\]](#)
- [5] Zhipeng Tang, Benlu Wang, Zhenyu Zhang, **Wuao Liu**, Pengda Xiang, Juncheng Huang, Chin Siang Ong, Hong Yu, Zonghai Yao. *MediSketch: Test-Time Scaling for Multimodal Patient Education*. Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.
- [6] Filippas Bellos, Yayuan Li, **Wuao Liu**, Jason J. Corso. *Can Large Language Models Reason About Goal-Oriented Tasks?* Annual Meeting of the Association for Computational Linguistics (ACL) Workshop (SCALE-LLM), 2024. [\[paper\]](#)

Industry Experience

- 05/2026 – 08/2026 **Generative AI Research Intern**, *LinkedIn*, Mountain View, CA
- Developed a semantic matching model for member-job fit-level prediction, directly modeling member profiles and job posts for efficient relevance scoring.
 - Fine-tuned Qwen3-0.6B for multi-level relevance prediction, achieving **76.9%** multi-class accuracy and outperforming the existing semantic-oracle baseline by **12.9** percentage points.
 - Proposed a rationale-distillation framework that learns from structured qualification-level matching rationales while retaining label-only inference, further improving accuracy to **79.1%**.

05/2025 – 08/2025 **Research Scientist Intern**, *Honda Research Institute*, San Jose, CA

- Implemented spatial audio data generation and filtering pipelines for a large-scale dataset containing **872k** question-answer pairs.
- Designed a spatial audio encoder jointly optimized for sound classification and direction-of-arrival estimation tasks.
- Developed a multimodal localization approach for indoor environments, integrating spatial audio understanding with large language models.

Research Experience

01/2025 – 12/2025 **Self-supervised Learning for Natural Sounds**, *University of Massachusetts Amherst*

Advisor: Prof. Grant Van Horn

- Extended the masked autoencoder (MAE) pretraining framework to bioacoustics.
- Conducted a systematic study of MAE pretraining on the iNatSounds dataset, analyzing the effects of pretraining data scale, domain specificity, data curation, and transfer learning.
- Enabled self-supervised representation learning via audio spectrogram reconstruction and achieved **62.76%** top-1 accuracy for species classification.

06/2023 – 04/2024 **Procedure Understanding from Egocentric Videos**, *University of Michigan, Ann Arbor*

Advisor: Prof. Jason J. Corso

- Designed a video collection protocol and curated a dataset of **224** cooking videos.
- Trained a Faster R-CNN object detector for **14** kitchen items, achieving **0.86** mAP on the curated video dataset.
- Developed a procedural step classification model achieving **92.54%** accuracy and enabled automated detection of step-transition errors.

Honors and Awards

2024 UMass Amherst Computer Science Ph.D. Fellowship

2021 Outstanding Graduate of Zhejiang University

2020 Xiaomi Scholarship

2020 Chunzhen International Exchange Scholarship

Professional Service

PC Member New England Computer Vision Workshop (NECV) 2025

Reviewer International Journal of Computer Vision (IJCV)

Technical Skills

Programming Python, C/C++, MATLAB, $\text{\LaTeX}$, ROS

Deep Learning PyTorch, TensorFlow 2

LLM Training Verl, TRL, Hugging Face ecosystem